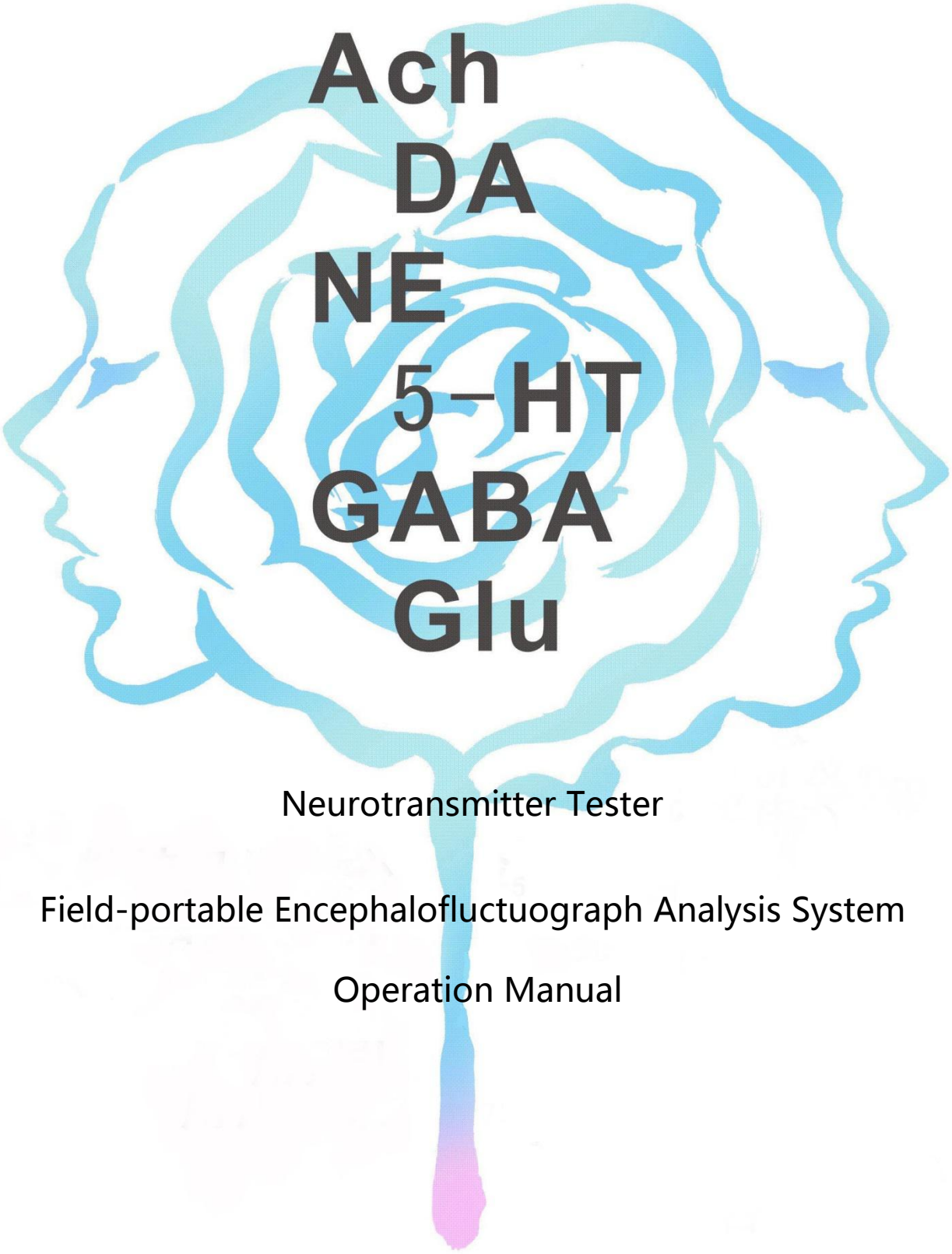

Neurotransmitter Tester

Field-portable Encephalofluctuograph Analysis System

Operation Manual

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Overview

Neurotransmitter tester, also known as the Encephalofluctuograph Technology (or ET), is a prospective EEG analyzer(made by Beijing Old Tong Ren Photoelectric Technology Co., Ltd), which can be used in the fields of psychological diseases, neurological diseases, brain injury, brain function selection of special person, evaluation of psychological consultation and treatment, intelligence development etc.

Its characteristics are:

1. One-time non-invasive extraction of six activeindex of neurochemical transmitter including GABA (Y-aminobutyric acid), GLu (glutamic acid), 5-HT (5-HT), ACh (acetylcholine), DA (dopamine), and NE (norepinephrine);

2.Judging the equilibrium state of neurochemical transmitters;

3.Assessment of brain function state;

4.Entropy assessment;

5.Intelligence assessment.

Clinical studies have shown that mental and psychiatricillnesses are one of the typical areas of neurotransmitter imbalance in the brain. Neurotransmitter tester (brain ET) can help clinicians to explore the etiology,thelapse to,the prevention and treatment of mental illness according to the biological indicators of neurochemical transmitters, changing the traditional diagnosis and treatment model of mental illness without objective index.

A detailed discussion on Encephalofluctuograph Technology in*ET-Brain Function Study*published by National Defense Industry Press in 1997.

Chapter Ⅰ Hardware

I. Hardware Configuration

1. Laptop (pre-installed software package)
2. 12-Channel EEG amplifier
3. USB communication adapter cable
4. 12Lead-in EEG combined electrode line (with inlet electrode paste)
5. Sterling silver saddle electrode nail
6. Brain electrode cap
7. Special trolley

II. Hardware Installation:

1. Connect the trapezoidal interface in the 12-channel EEG amplifier with the 12 lead-in connector;

2. Connect the USB interface of the 12-channel EEG amplifier to any USB interface of the laptop.

Chapter Ⅱ Software

I. Installation and Uninstallation

1. Install the setup file in the Drive folder of the U disk, drive the installation: click the SETUP file in the "Drive" file of the U disk;
2. Software installation: place the included software installation U diskinto USB, and click the setup file in the "Program Installation" file of U disk to install; after the installation, the software will automatically establish a shortcut called "monET" on the desktop;
3. Uninstallation: In Control Panel, select monET and click Uninstall to completely remove the software;
4. If the user selects self-installation, the STD.datin the Calibration Files folder on the CD needs to be copied to the DATA folder after the installation. (the STD file is the signal file for portable ET amplifier)

II. Notes and Precautions

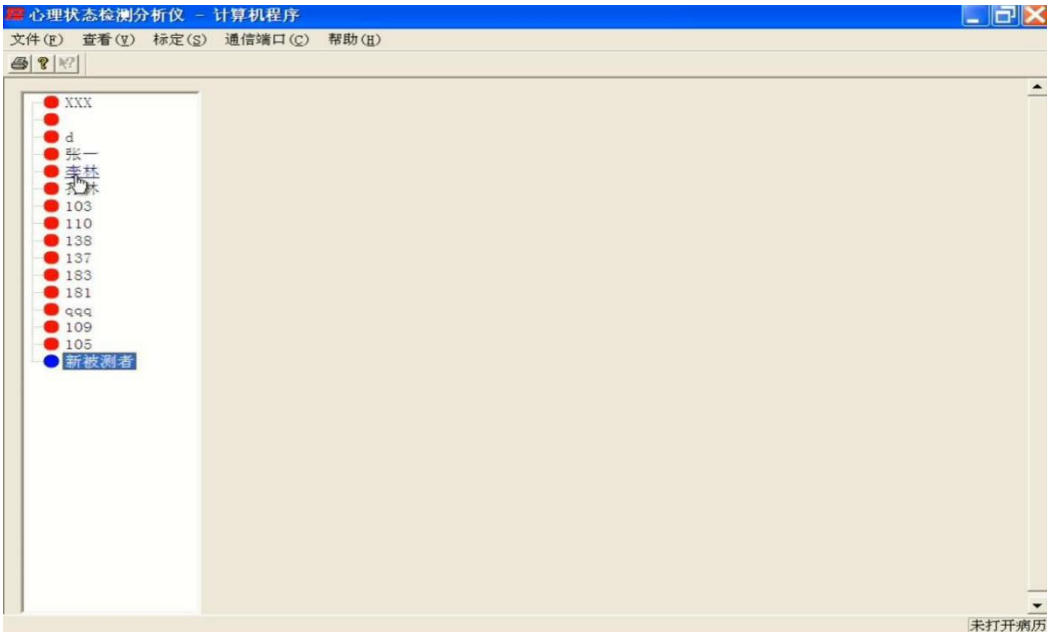
1. The software has been pre-installed in the attached computer, so the user only needs to connect the computer with the 12-lead EEG amplifier.
2. The DATA folderin the software installation directory,all the testers' data, is very important (default installation location; if you need to restore your system or reinstall the software, copy the folder to a different partition on your DC, and after the operation, copy the folder again where you reinstalled the software;
3. The resolution of the computer is 1024 * 768, and the display font will be adjusted to Big Font; (in the pre-installed computers, all have been adjusted to the situation required by the software, so you need not to adjust);

III. Instructions for the Use of Software:

1. Click the shortcut "monET" to enter the software (as shown in Figure 1):

(Figure 1)

2. Enter the interface of the software (as shown in Figure 2):



(Figure 2)

3. File menu (as shown in Figure 3)

(Figure 3)

Click File menu and the four sub-menus including Using Unit, Print, Print Settings and All Print appear (Figure 4).

(Figure 4)

1). Using Unit-modify the name of the using unit when setting the print report;

2). Automated Assessment-once selected,the printed report will include analysis software's Automatic analysis for the testee;

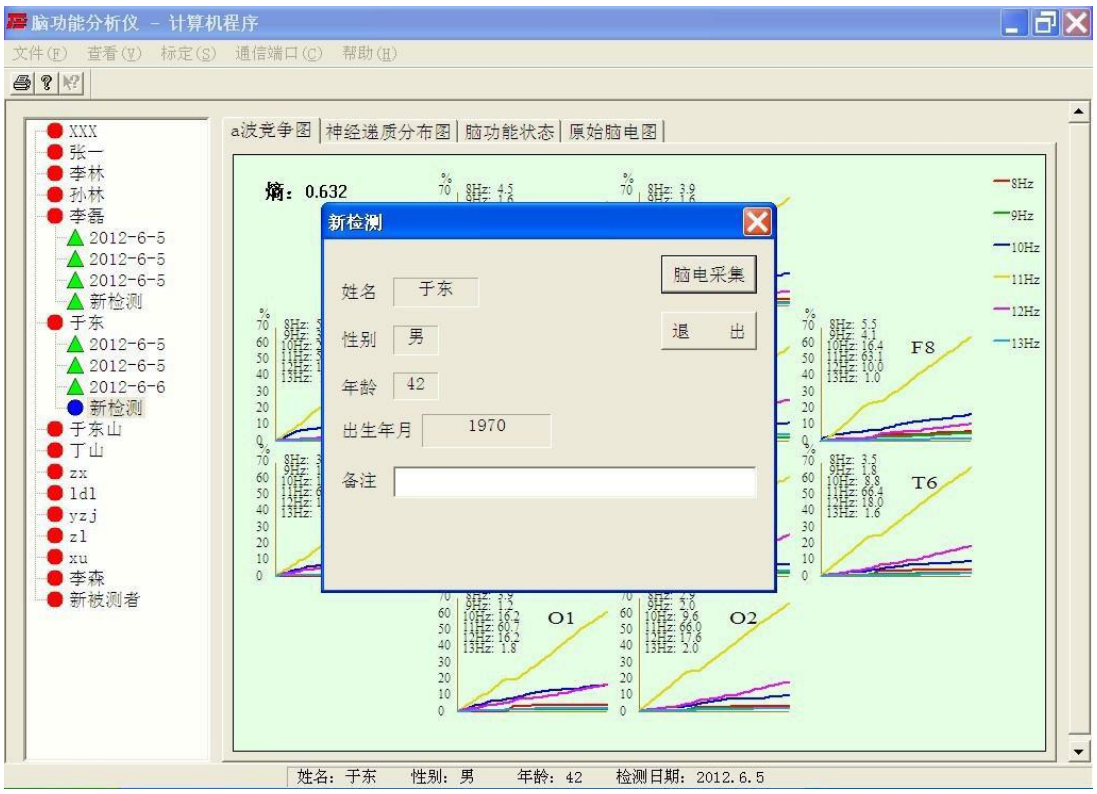
3). Print Settings - select a printer and printer connections;

4). Print - print the current page;

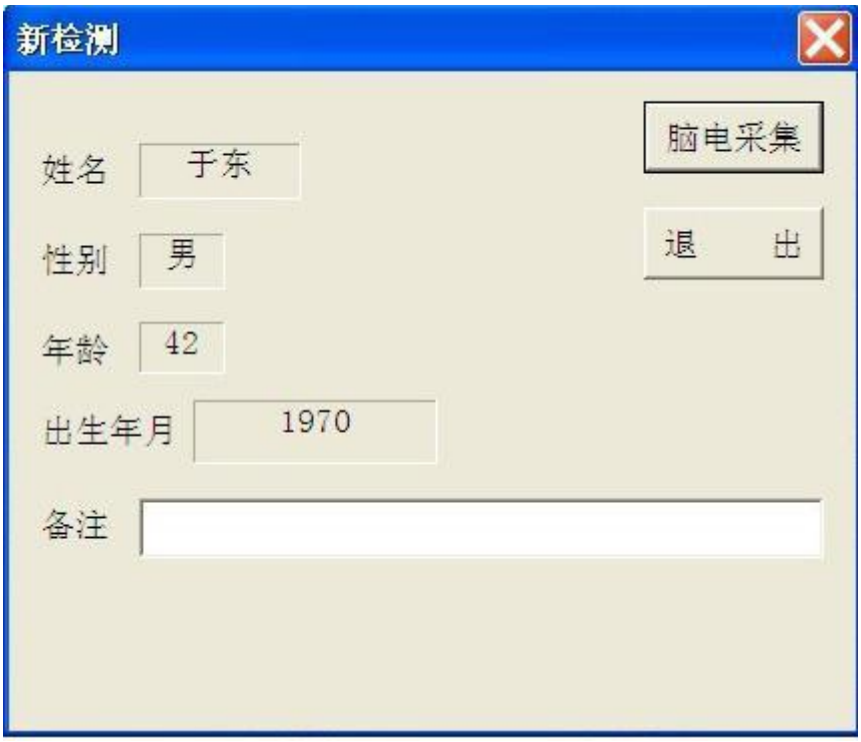
5). All Print- a full report of one test for the currently selected testee;

6.) The "Calibration" and "Communication Ports" in the File menu do not require user to set.

4. Medical Record Information Bar:(left in FIG. 5) is a medical record display field that includes information of all the testees. After clicking, enter the sub-menu for continue operation (as shown in Figure 6);



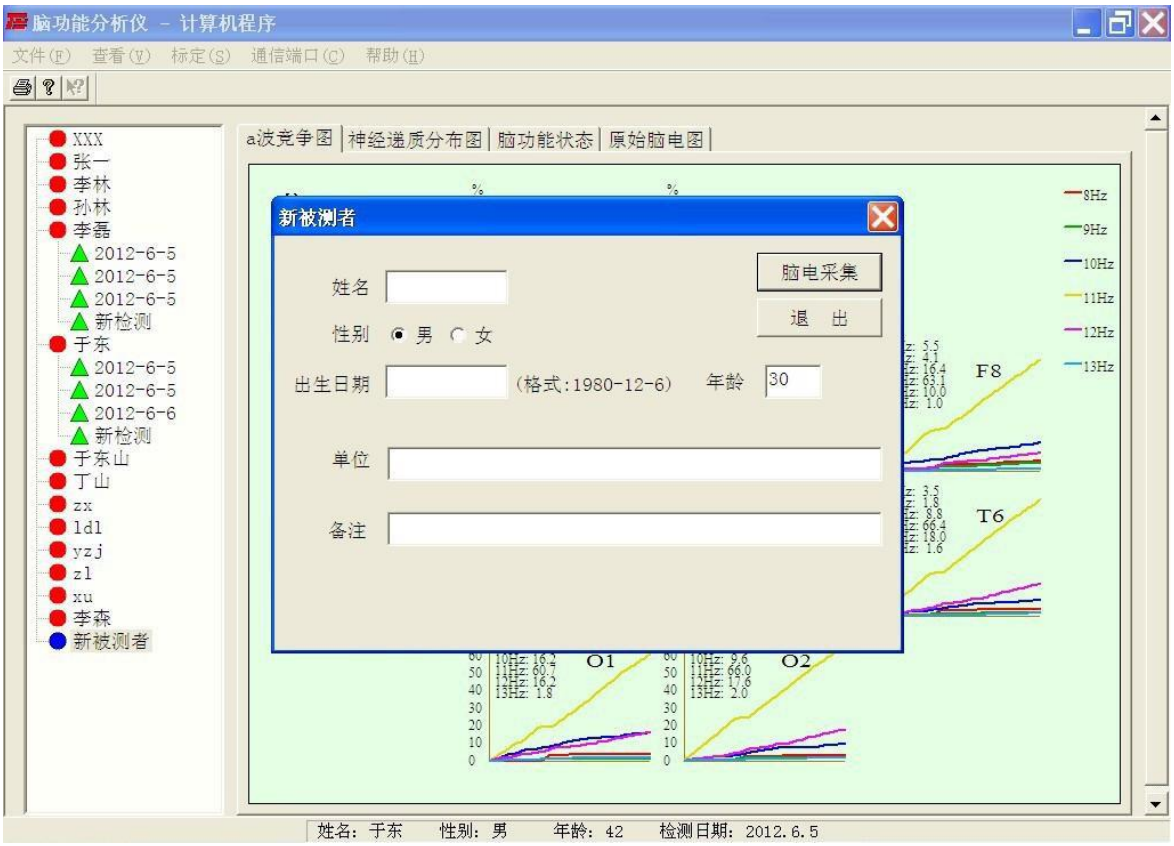
(Figure 5)



(Figure 6)

Here, you can select the existing testees to view the test results or perform a new test (Figure 5, 6, and 7). You can also click the a new testee to perform a new test (as shown in Figure 8);

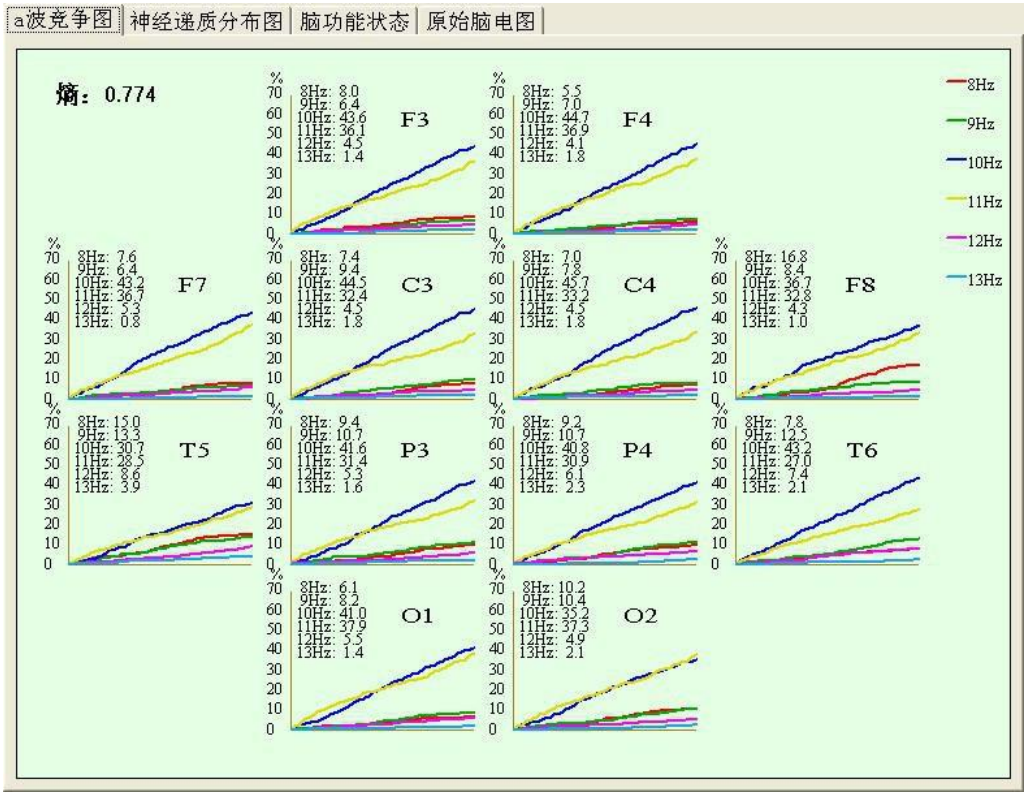
(Figure 7)



(Figure 8)

5. Click on a certain test of a testee to enter the focus of brain function analysis -- data analysis, whichis divided into four categories:α-wave competition map, neurotransmitter profile, brain function state, original electroencephalogram

① α-wave competition map(as shown in Figure 9)：



(Figure 9)

EEG not only has power fluctuation, but also has frequency fluctuation, which shows the continuous transformation between different frequencies. The α-wave competition graph shows all six frequencies of the α-wave (8 Hz-13 Hz in the upper right) predominate in the α segment.

When the probability of a certain frequency is significantly higher, the probability of other frequencies is significantly reduced.At this time, the order degree of the brain is the largest, and the entropy value is the smallest.On the contrary, when the probability of six frequencies is close, their curves are basically regrouped together, the order degree of the brain is the smallest, and the entropy value is the largest.

The normal entropy value shall be at 0.45 to, 0.85.

Outside the range is abnormal, and too high entropy value indicates that the system is mixed and that the outside information cannot be accepted or processed. And the same for too low entropy value with closed state.

We can simply consider the dominant frequency as being related to intelligence, and the frequency (8 Hz to 13 Hz) is proportional to the level of intelligence;

② Neurotransmitter profile (Figure 10):

(Figure 10)

Figure 10 shows the activity and distribution of six neurotransmitters and two receptors in the brain including GABA (Y-aminobutyric acid), GLu (Glu), 5-HT (5-HT), ACh (acetylcholine), DA (dopamine), NE (norepinephrine), S3-lineage ACH receptor, and S13-lineage deep inhibitory receptors.

Normal value: normal mode value established by clinical trial of standardized sample and representative sample;

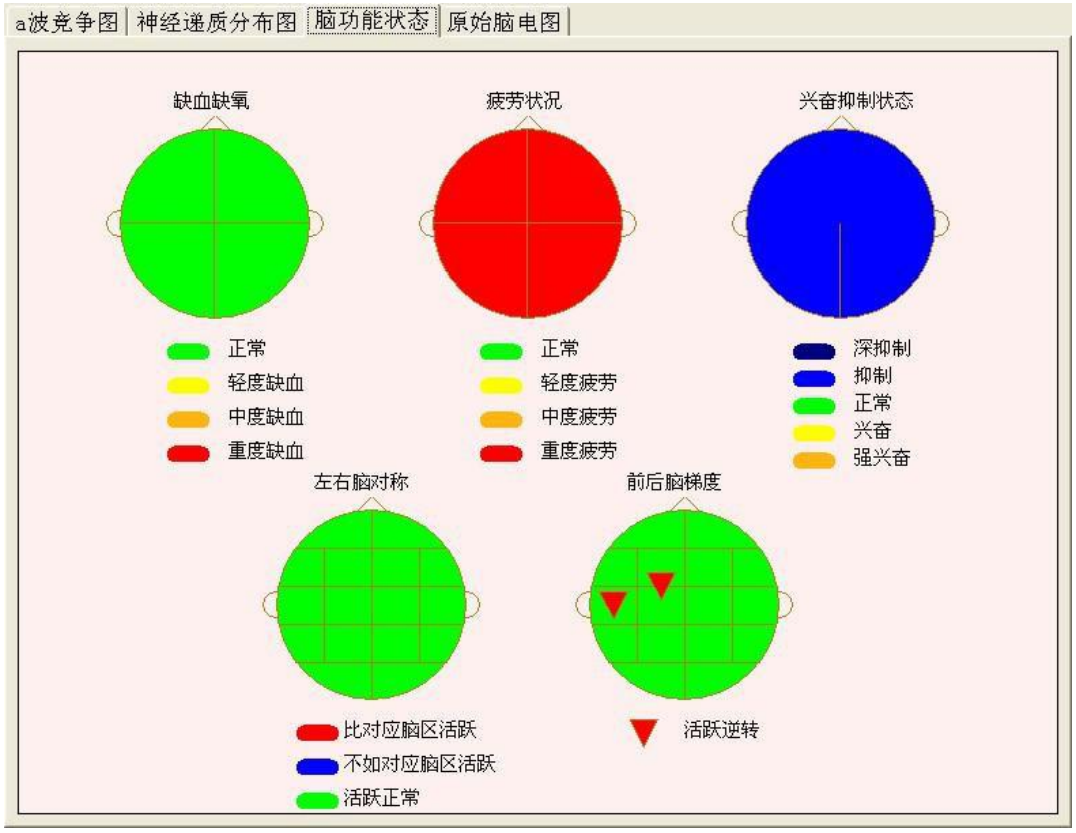
Measured value: indicate the test data of the testee;

Distribution of brain regions: the distribution of the measured values in the four brain regions of anterior left brain, posterior left brain, theanterior right brain, and posterior right brain.

The red and green bar graph represents the normal values and measured values of the current six neurotransmitters and two receptors. The green part is the normal value, and the red part is the measured value. Deviating from the normal value indicates the symptom of the body, psychology, behavior, feeling, movement and function of the testees.

(This analysis software contains normal mode with five years as an age, so please ensure the accuracy when filling out the age of the testee. This norm is provided by the authoritative organization with assured reliability.)

③ Brain function state (as shown in FIG. 11): The brain function state is the evaluation of brain function in the analysis software, and the function state of each brain region is represented by different colors, includingischemia/ hypoxia, fatigue condition, excitation inhibition state, left and right brain symmetry, anteroposterior cerebral gradient;



(Figure 11)

Ischemia/ Hypoxia

In animal trials, structural change of S1...S8 and S23, S27, S28, S29 and other spectral lines and harmonic frequency enhancement appear when brain blood supply disorders. Animal ischemia results from suspension test can cause animal blood vessel obstruction, and the same when human brain blood supply disorders.When disorder appears above 10 times in the whole brain, that is, the whole brain was in the ischemic state; the half-brain more than 5 times, that is, the half-brain was in the ischemic state; and the 1/ 4 brain region over 3 times, that is, the 1/ 4 brain was considered to be in the ischemic state.

Fatigue condition

The spectral lines entering the dominant frequency region sometimes have a continuous phenomenon, such as S2, S3, and S4.We call this a continuous frequency that can be divided into two-frequency, three-frequency and four-frequency.The shorter frequency, such as two-frequency, may reflect the decrease of the single frequency selectivity, or the increase of a certain frequency bandwidth, while the longer continuous frequency a manifestationtrending to chaos.As we observed in memory tests that the continuous frequency increased at rest while decreasing significantly at memory time, indicating that the increase in the intensity of the continuous frequency was associated with the decreasing brain activity.In the blood supply disorders associated with the basilar artery system, there are often more continuous frequencies in the posterior part of the brain, and sometimes the position where the pathological state is located can be distinguished.

Excitation inhibition

Stimulation and inhibition are the basic dynamic processes of the nervous system. Their coordination is such an important condition that the population can adapt to the external environment, and the change of external stimulation conditions can induce the changes of excitation and suppression activities.

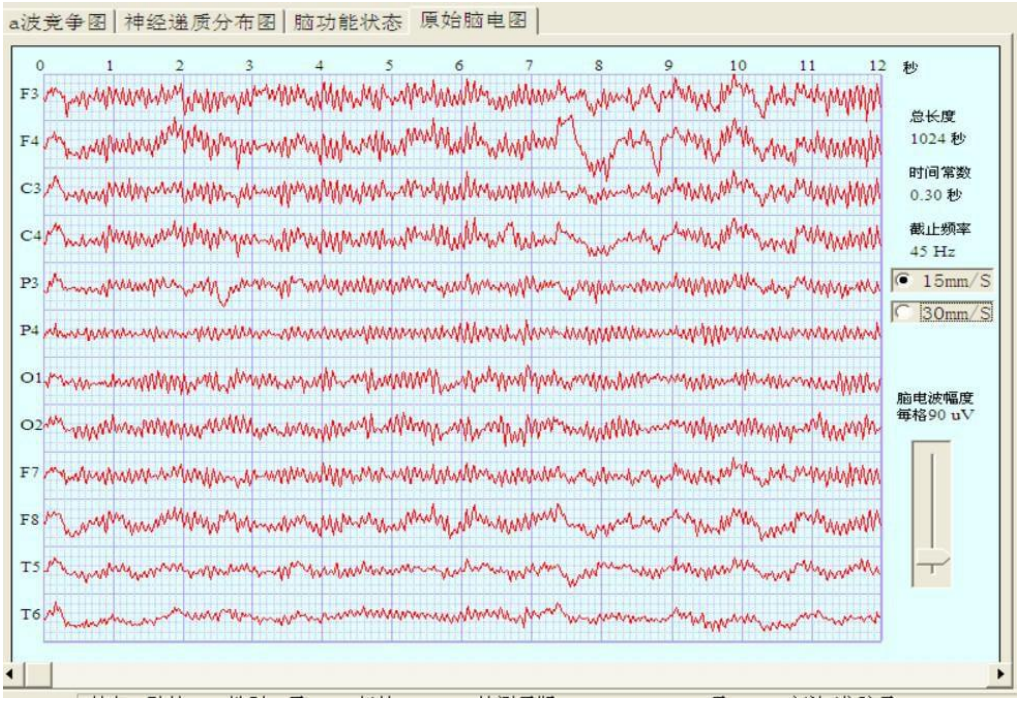
S1 (γ-aminobutyric acid) is an inhibitory amino acid, and S2 (glutamic acid) is an excitatory amino acid. Therefore, they are antagonistic but act together to maintain the balance of excitation and inhibition of the brain.Thus, when such a balance is broken, there will be a variety of lesions.

Symmetry of left and right hemispheres

Whether the power of all six frequencies (8 Hz-13 Hz) of the α-wave is symmetrical or not is displayed in 12 brain regions, respectively; **the anteroposterior cerebral gradient:**

The average power of each brain region varies in gradient before the age of 40, that is, increasing year by year. If the power relationship of left and right brain andforebrain and hindbrain all obey the golden rule, thebrain image structure is probably the best.With the increase of the age, the general trend of power distribution in the brain area after 40-year-old is that the location of the maximum power moves forward. The rule that thepulvinar (especially 02) causing other brain regions disappears. The average power distribution is consistent with the whole brain, moreover, the average power distribution of the whole brain tends to be consistent leading to themonotonicity and inhibition of the brain activity,and the changes in the forebrain, especially the left forebrain, are most pronounced. The study demonstrates that the role of the left forebrain is the brain region that governs memory and intellectual activity.

④ Original electroencephalogram (Figure 12):



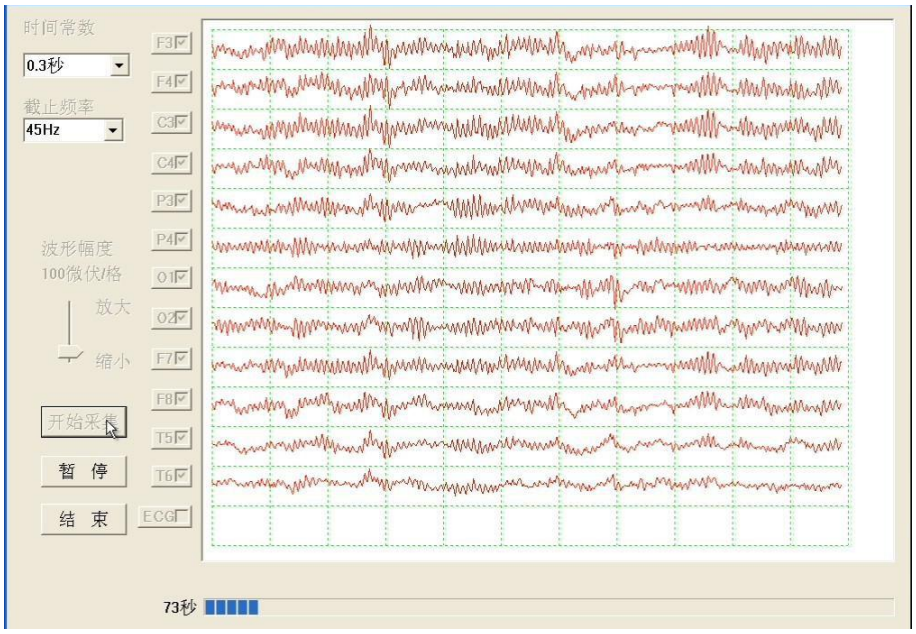
(Figure 12)

The original EEG is a real-time recording of the EEG. It is the same as the general paperless EEG. There are two index display methods: 15mm / S (12 seconds per screen) and 30 mm/S (6 seconds perscreen);scroll bars can be moved per second or moved anywhere;

6. How to test: click the name in the information area of testee (Figure 6) or click “new testee” if the new testee needs to be created (Figure 7), fill the information of testee, click EEG collection and enter the test page.

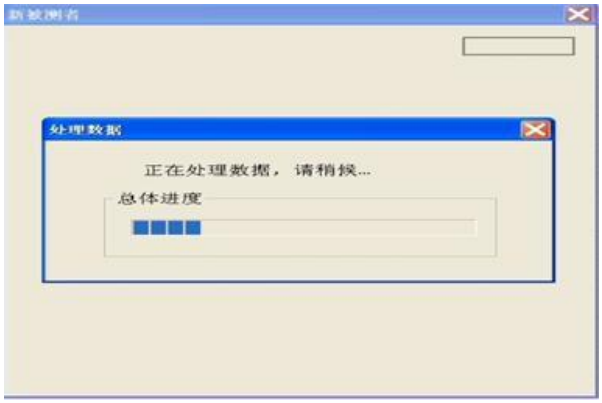
After entry, the frequency collected by ET amplifier will be displayed in waveform. Please check whether the information collected by the amplifier is pure and undisturbed first to ensure thatthe waveform (Figure 13) is correct. The correctwaveform should be exquisite and regular α wave and fluctuate forward in the green block. There is the name of theelectrode in the corresponding brain region on the left side of each waveform. If the middle box is checked, it means displaying the waveform in the current brain region;otherwise, it means not displaying.

Choose 0.3 second for “time constant”, 45 HZ for the “cut-off frequency” and 100 microvolt/grid for the waveform range on the left side.



(Figure 13)

Confirm that there is no interference waveform in each leads, and then click the start collection and the EEG will be collected. The collection will last 18 minutes and then the software will automatically save the result. (See Figure 14)



(Figure 14)

After the software returns, select this test of the testee in the information bar of case history and check, analyze and print the test (including the analysis report).

Note: automatic analysis report given by analysis software can be seen only when allreports of thetesteeare printed (select “print all” and “automatic evaluation”in the file menu as shown in Figure 4).

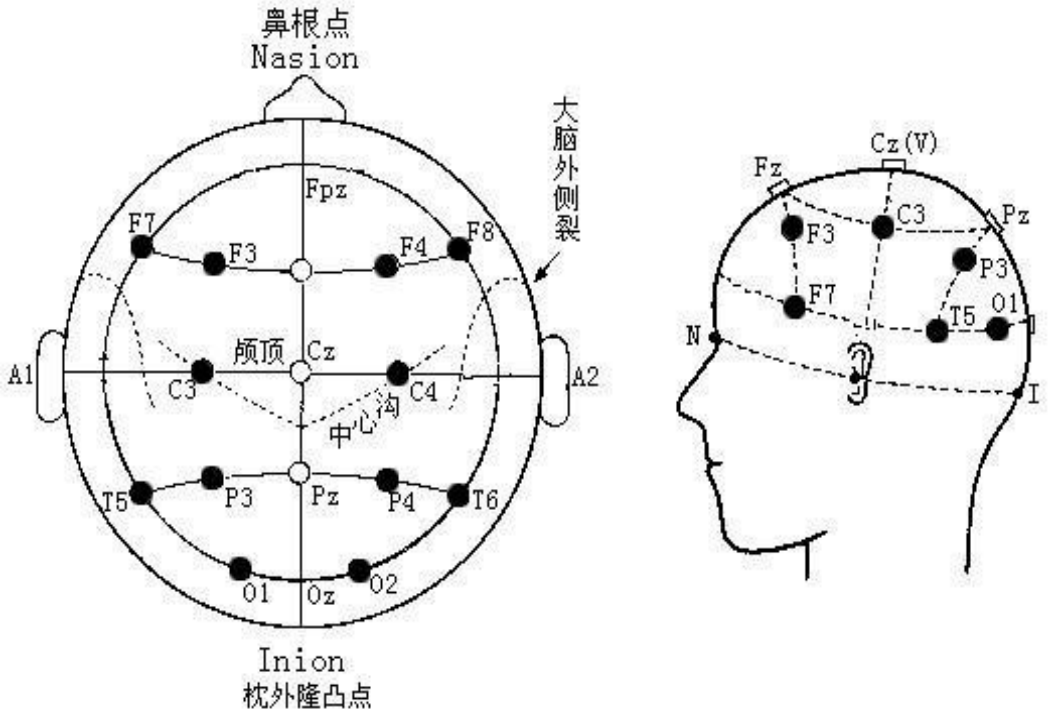
Chapter Ⅲ Instructionsto Wearing

I. The testeeshall wear the brain electrode cap;

For the accompanied electrode cap, which has been basically divided into five parts by the back-to-back connection of rubber tube as per the shape of brain region, the user just needs to adjust the tightness based on the head sizeof thetestee.

II. The placement of the electrode;

Place the electrode in accordance with the 10-20 system of electrode placement of the international standard lead: set the connection line between the nasion and the inion as the middle line, set the connection line across the middle line between the two prefossettes of two ears as the transverse line, place the electrode separately at the 10% part of the middle line and 20% part of the transverse line, carefully part the hair before placing the saddle electrode with fine silver, make sure the electrode nail fully touching with the scalp and degrease the scalp with 95% alcohol. (See Figure 15)



大脑外侧裂	Fissure of sylvius
颅顶	Calvarium
中心沟	Central sulcas

(Figure 15)

III. Each lead of 12-lead EEG compound wire has marked the corresponding connection place (**theodd number stands for the left side, the even number stands for the right side, L stands for the electrode of left ear, R stands for the electrode of right ear and GND stands for the ground electrode**), and it is only required to connect the crocodile clip of the wire head with the electrode.

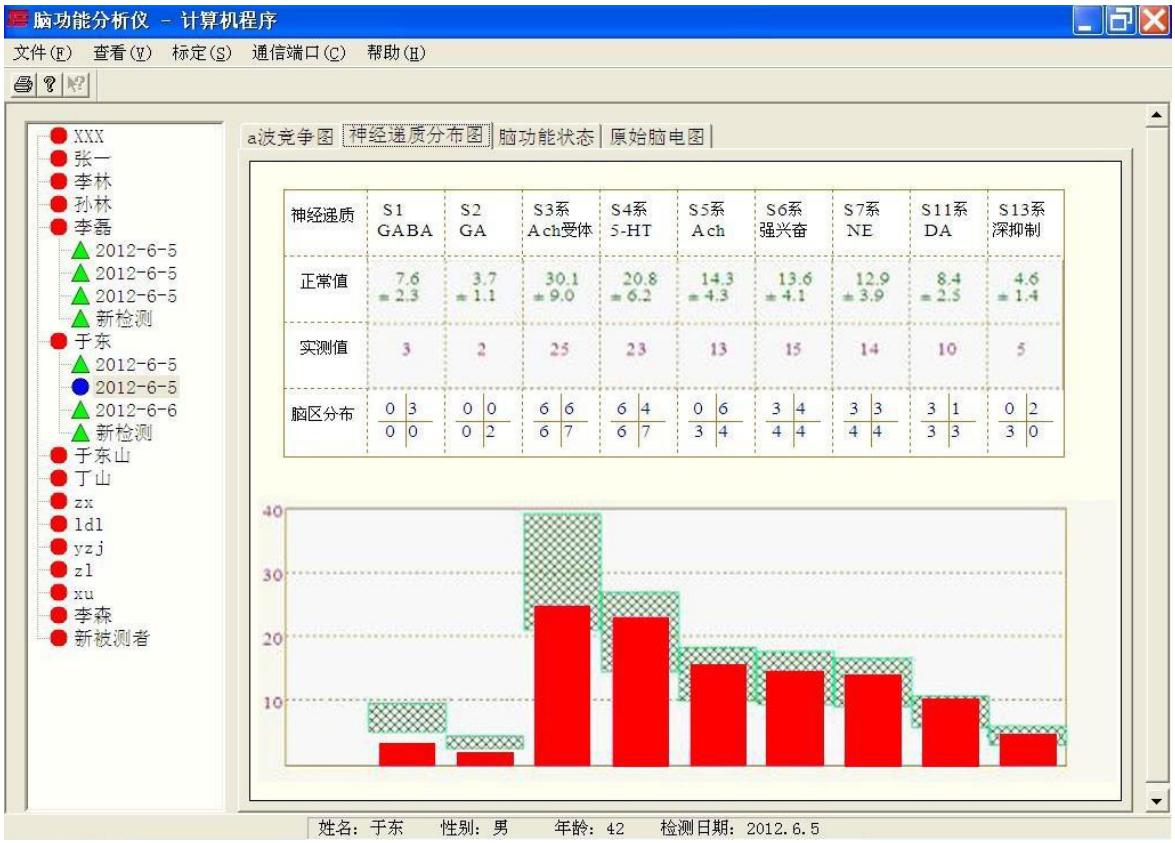
IV. Apply evenly the electrode jelly in the electrodes of both ears of the 12-lead EEG compound wire and clip them onto the two earlobes separately as per the mark.

V. With regard to the placement of electrode, please watch the video attached with the device.

Chapter Ⅳ Interpretation of the Case History

In addition to their physiological function, the neurotransmitters in the brain also restrict each other and keep the function of the brain in a balanced state. Once the balance is broken, the brain becomes dysfunctional. Therefore, the equalization of neurochemical transmitters directly reflects the human life quality.

I. ET Characteristics of Insomnia:



(Figure 16)

The decrease of S1 and (γ-aminobutyric acid) and S2 (glutamicacid) inside the testee reminds the abnormal activity of GABA and GA in the brain. The inhibitory effect of GABA on the central neuron was reduced and GA was not sufficient to stimulate the sleep center.

II. ET Characteristics of Cephalagia(Figure 17)

(Figure 17)

It was shown that GABA (γ-aminobutyric acid) and GA (glutamic acid) increased significantly, illustrating that the two neurotransmitter activities increased significantly, and the activity of GABA (γ-aminobutyric acid) increased during the intermittent period, which obviously manifested thecharacteristics of cephalagia.

III. ET Characteristics ofPsychiatric Symptoms (Figure 18)

(Figure 18)

Infigure18, thetestee presented with multiple changes in mood, showing a marked increase in the glutamic acid (related to ischemia, excitation, head pain, maintain an excitatory fatigue state); a significantdecrease in dopamine, a significant increase in S1 GABA (γ -aminobutyric acid); and a significant decrease in S7 NE (Norepinephrine), whichconform to the activity characteristics of the neurotransmitter with depression.

Chapter Ⅴ Interrelationship and Interactions of Neurotransmitter

I. The main function of central neurotransmitters and the relationship between their dysfunction and morbid state:

1.γ-Aminobutyric Acid (GABA):it belongs to the inhibitory amino acid and has the physiologicalactions such as antianxiety and inhibition of hypothalamo-hypophysealsystem. It will keep the balance of excitation and inhibition of brain with the glutamic acid and help the brain select or reject the outside information with the selection function.

The increase in gamma-aminobutyric acid is common seen in the administration of diazepam, where the brain is in an inhibitory state and the response to the outside is impaired. The increase of γ-aminobutyric acid is also found in the inhibitory state of the brain, namely the aging state.

The decrease of γ-aminobutyric acid is usually found in epileptic seizure and convulsive patients. At this time, the antagonism of glutamate is out of balance, the effect of glutamic acid is dominant, and the brain is in an excitatory state. In clinical observation, the attenuation of γ-aminobutyric acid not only appeared in the intellectual disability but also in the state of cerebral ischemia.

2. Glutamic Acid (Glu): It belongs to an excitatory amino acid to maintain the excitability of brain, which is the basis of all activities in the brain. The decrease of glutamic acid can lead to decreased function of cerebral cortex, memory loss, aprosexia, weakness of exercise ability, etc., the decrease of glutamaic acidcan also lead to the decrease of the control regulation function of the lower parts of the cortex, which can easily cause the disorder of the brain function. The decrease ofglutamaic acidis common seen in fatigue, drowsiness, etc., and the glutamic acid is extremely reducedin anesthesia and coma.

The glutamic acid value in the patients with insomnia was obviously increased, and the content of glutamic acid in the brain of epileptic patients was obviously increased, but the γ-aminobutyric acid decreased significantly. It is considered that this is one of the neurochemical bases for the abnormal excitation and the onset of epilepsy, and the phenomenon is not shown after the onset. Therefore, the glutamic acidhas an essential connection with theepilepsy, and it is the trigger factor of epileptic seizure. In clinical practicetheusage of barbiturates is for the treatment of certain epileptic seizures.

The glutamic acid and γ-aminobutyric acid are a pair of opposite in function and functionally antagonized transmitters that act together to maintain the balance of the excitation and inhibition of brain, and when this balance is broken, various lesions will arise. Overexpression of theglutamic acid was the same as low expression of γ-aminobutyric acid, while too low function of glutamic acid is same as too high function of γ-aminobutyric acid. The excess of the glutamic acid may also cause damage to brain cells, (such as the reperfusion injuryin ischemia).

The decrease of the γ-aminobutyric acidand the glutamic acid will happen at the same time in patients with cerebral fatigue, children's hyperactivity and some mental disorders. The mechanism may be that the decrease of the γ-aminobutyric acidandtheglutamic acid leads to decreased function of cerebral cortex, and the decreased control of subcortical tissue, and moreover the function of subcortical tissue is disturbed, hence the various clinical symptoms appear.

3. 5-Hydroxytryptamine (5-HT): It mainly reflects the fatigue state of the brain center. It also plays an important role in keeping the mental stability and regulating the functional balance of neurotransmitters in the lower parts of the cortex. The insufficiency of 5-hydroxytryptamine may cause emotional disorders; and the decrease of5-hydroxytryptamine and norepinephrine may cause depression. The increase of 5-hydroxytryptamine appears in anxiety disorder, children hyperactivity and other diseases.

The 5-hydroxytryptamine in patients with affective disorders will decrease. The decrease of 5-hydroxytryptamine may result in a mental imbalance. It may be associated with the extent of norepinephrine at the same time as development of manic or depression. In general, the increase of norepinephrine may cause a rise in emotion or the like. 5-Hydroxytryptamine had different changes in patients with the migraine, the cerebral ischemia, the mental retardation and the Parkinson's disease. The decrease of 5-hydroxytryptamine can cause insomnia, butthe administration of 5-hydroxytryptamine (5-HTP) can return the sleep.

4. Acetylcholine (ACh): It is mainlyinvolved in regulating the excitatory activity of the brain, such as learning, memory, sleep, exercise, eating, and regulation of body temperature.

The increase of acetylcholine is easy to form an excitation focus in the brain andlead to blood pressure rise, dyskinesia, and other symptoms. High blood pressures, Parkinson’s diseases, epilepsies and so on are common diseases. In patients with Parkinson's disease, thedopamine decreases while the acetylcholine increases. It appears that the acetylcholine is excessive and is treated with an anti-acetylcholine drug (antan) to restore it to normal. In the patients with vertigo, the cholinergic structure of the limbic hypothalamus system was abnormal, and the acetylcholine increased, and the anticholinergic drugs were often used to treat vertigo. The decrease of acetylcholine causes cognitive impairment, theintelligencedecrease and common disease such as Alzheimer's disease. There was a significant decrease of acetylcholine in children with mental retardation; acetylcholine attenuation in Alzheimer's disease; and abnormalactivity of acetylcholine in children with myasthenia gravis.

5. Dopamine (DA): The role of dopamine (DA) is to stimulate the functional state of the brain, regulate body movements, and participate in emotional activities. In exercise control, the role of dopamine is to regulate the degree of muscle tension, help the body ready for exercise, and "start" a certain action under the signal triggering of the cerebral cortex.

The increase of the dopamine leads toschizophrenias, chorea, and drug-seeking behavior (drug addiction). The dopamine in the brain of patients with schizophrenia will increase dominantly in the frontal and left brain. Treatment with perphenazinecan reduce it. The increased dopamine activity can be seen in patients with hyperactivity.

The decrease of the dopamine causes acute upper gastrointestinal bleeding, facial apathy, dyskinesia, and the like (e.g.Parkinson's disease). The dopamine neurotransmitters in patients with primary Parkinson's disease are significantly reduced. The patients with dysgraphia can see a decrease in dopamine activity in the cheirokinestheticcenter F3.

6. Norepinephrine (NE): Its physiological function is to control emotion. It has the vasoconstrictioneffect and regulation of exercise function. It co-assisted the dopamine system and controls the movement of the engine body on the basis of dopamine.

The increase of norepinephrine may cause cerebral vasodilation, hyperactivity headache, migraine, etc.; if the 5-hydroxytryptamine function is reduced at the same time, the mania may occur. In the state of cerebral ischemia, there will also appear the norepinephrine in the accumulation of excitation media. The elevation of S7 was observed in patients with cerebrovascular disease.

The decrease of norepinephrine and the concomitant weakness of 5-hydroxytryptaminewill cause the depression. The norepinephrine decreases and the dopamine increases in the patients with schizophrenia, (this change is more pronounced in the front and left of the brain, and the treatment with perphenazinecan make it decrease); the norepinephrine in the patients with depression is significantly reduced, suggesting that norepinephrine activity is weakened, whilethe adrenaline increased in the brain of patientswho turns into mania.

II. In addition to their physiological functions, the neurotransmitters interact with each other and keepthe brain function in balance.

Once the balance is broken, the brain becomes dysfunctional. There are three major functional pairs of transmitters in the brain: the gamma-aminobutyric acid and the glutamic acid, the dopamine and thenorepinephrine/acetylcholine and the 5-hydroxytryptamine,

the 5-hydroxytryptamine/norepinephrine. The 5-hydroxytryptamine can also regulate all transmitters, and it can regulate and maintain the balance of brain functionwhen other transmitters are abnormal.

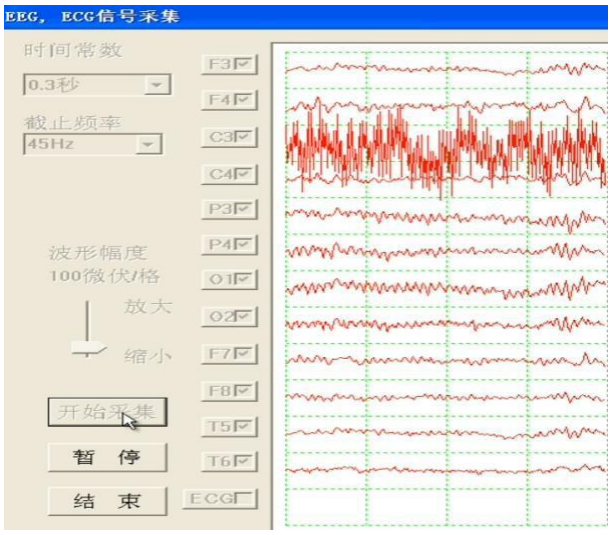
GLu and GABA arethetransmitters with adverse effect and antergic function, and they work together to maintain the balance between excitement and inhibition of the brain, or otherwise, the brain disorder will occur.

It is known that four neurotransmitters like ACh, 5-HT, DA, and NEcontribute to controlling the excerise. The r AChplays a role to maintain muscle tension; DA is to adjust muscle tension, so that the body canprepare for the movement, and in the cerebral cortex signal trigger the "start" of an action; and NE is to carry out a series of actions of the bodybased on DA; 5-HT is to inhibitDA and NE energy system. The interrelation among these four neurotransmitters is that: the DA and NE cooperate with each other; DA and NE inhibit ACh; 5-HT inhibits DA and NE, so 5-HT indirectly enhances the function of ACh which forms an indirect synergistic relationship. Ach and 5-HT in the regulation of exercise play a role of inhibition, and DA and NE play a role of excitement. These neurotransmitters work together to maintain the balance of exercise function of the brain.

5-HT and NE can also regulate the expansion and contraction of the blood vessel of the brain, but they are of adverse effect. 5-HT contracts the blood vessel of the brain and NE expands the blood vessel of the brain.

Chapter Ⅵ Common Questions and Solutions

Ⅰ.Identification and Solution for the Poor Single Electrode Signal

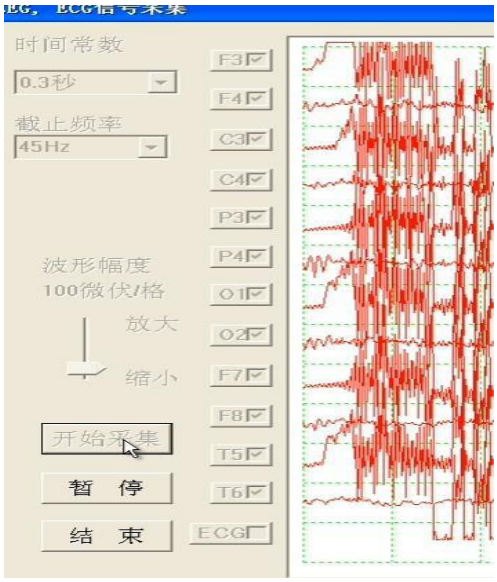


(Figure 19)

In figure 19, an electrode wave shows the solution to disorder(figure 19 shows C3 electrode picking up wave disorder)

1. Determine the name of the electrode first (letter and number on the left represent the name of the electrode), and then check whether the connection between the subject and the electrode is loose.
2. If the first condition is excluded, it must be the poor contact between scalp and the electrode. Please degrease the scalp with 95% alcohol and make sure the scalp and the electrode fully contact with each other.
3. If it is found that some poor electrode during the test, in order to make sure the test is accurate and effective, please press the “Pause”, and then press the “start collecting” to recollect after adjustment.

Ⅱ Connection of the Ear Electrode



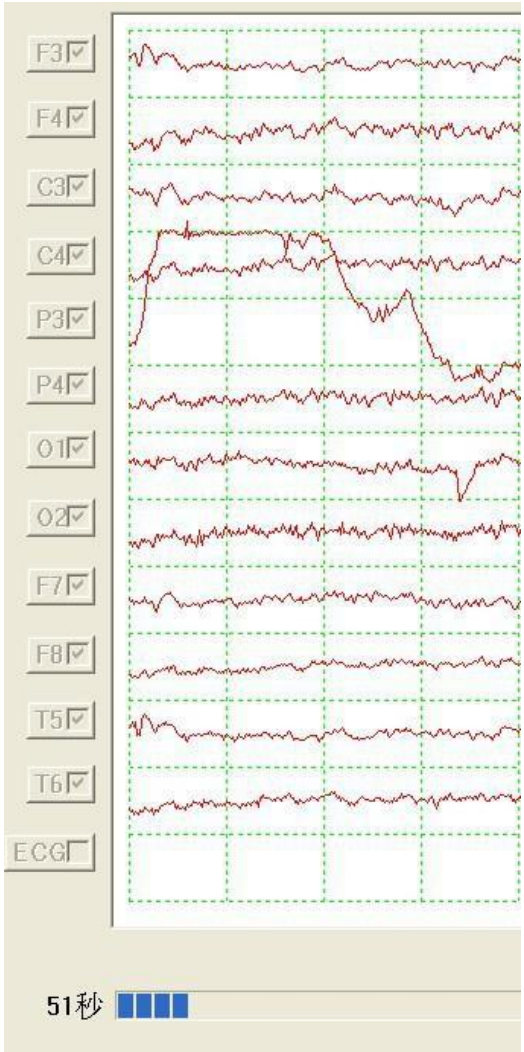
(Figure 20)

In figure 20, the wave shows the odd electrode disorder. It must be the poor contact of the left ear electrode and solutions:

1. Check whether the electrode paste in left ear electrode is sprayed uniform and flat;
2. Degrease the part of left ear that contact with the electrode with 95% alcohol.

III Some Electrode Wave in Float Display:

The wave picked up by P3 electrode in figure 21 displays in irregular float pattern. This must be having placed the electrode on the blood vessel of the brain when placing it. Please press “Pause”, and place the electrode nail again;



(Figure 21)

Chapter Ⅶ Precautions

Ⅰ. ET analysis software can only be operated under **single-core CPU, 32-bit** operating system;(pre-loaded laptop and have already installed operating system, analysis software and do not need to set), the following are the steps to close multi-core CPU merely for reference when you debug again:

1. Click operating in start menu, enter “msconfig”and click enter;(the following figure take the XP system as an example)

(Figure 22)

2. Click BOOT.INI，and continue to click advanced options(Figure 22);

(Figure 23)

3. Select /NUMPROC in (Figure 23), and select 1 in the dropdown list and click confirmation.
4. Restart the computer.

Ⅱ. Before each use of pure silver electrode nail, use physiological saline to soak for 2 hours or more. After use, wash it with clean water and air-dry to prevent noise interference in the next use due to pollution; after use, 12-lead conductor in the left and right ears electrode, should be cleaned from applied electrode paste with a cotton swab;

Ⅲ. Analysis software needs the result obtained from calculating the measured value of the subject and the norm value in the software. Therefore, **the age of the subject must be filled in correctly. (The norm value in the software is provided by the authority and is reliable)**

Ⅳ. Two days before test, subjects cannot take the drug associated with encephalopathy; when testing, the test environment should be kept quiet, subjects should close the eyes, and in the course of the test cannot fall asleep, cannot blink or swallow, and can think of something pleasant;

Caution:

In order to ensure the 12-lead EEG amplifier can collect the purity ofultraslow EEG frequency, the power line of the laptop must be three phase power plug. The external power of the laptop must be three phase 220V reachctrl power that contact well with the ground. Otherwise, during the 18 minutes of EEG collection, the power line of the laptop should be unplugged and the laptop be powered by the accumulator. After the test, plug the power line again.

Chapter Ⅷ Maintenance, Storage, Package and Transportation

I. Maintenance

1. If the instrument is not working properly, please contact the manufacturer directly. The manufacturer is responsible for maintenance or giving technical guidance. Do not disassemble on your own, otherwise affecting the use on your own account.

2. When cleaning the stain on the instrument surface, confirm that it is powered off. First wipe with slightly wet medical gauze (it is strictly forbidden to let liquid flow into the instrument through the gap) and then use dry gauze to wipe. The LCD screen is forbidden to wipe with alcohol or other chemicals. And the instrument shell shall not be wiped and cleaned with alcohol, thinner or liquid containing acid or alkali.

3. When the instrument is suspended, cover it to prevent dust or moisture.

4. If the set of equipment reached service life it cannot be discarded and should be recycled according to the relevant hospital regulations.

5. When the signal collected by the instrument is abnormal, the common mode rejection ratio and frequency range are to be measured and calibrated by the manufacturer's maintenance engineer.

II. Storage

1. Temperature -20℃-55℃, relative humidity not more than 93%, atmospheric pressure 500HPa~1060 HPa non-corrosive gas and well ventilated indoorenvironment/warehouse.

2. Indoor environmentshould be clean with no insects, no mouse, and to ensure that the instrument is without damage.

3. Products should be placed alone, surrounded by gaps, easy to move and check, if necessary, clean the outside surface, replace the cover items.

4. When the instrument is suspended, cover it with cloth to prevent dust or moisture.

5. Long-term deactivation of the instrument:temperature -20℃-55℃, relative humidity not more than 93%, atmospheric pressure 500HPa~1060 HPa non-corrosive gas and well ventilated indoorenvironment/warehouse; indoor environmentshould be clean with no insects, no mouse, and to ensure that the instrument is without damage; products should be placed alone, surrounded by gaps, easy to move and check, if necessary, clean the outside surface, replace the cover items.

III. Package and Transportation

1. Outside the box there should be up and moisture-proof logo, and during the transport process it should be preventedfrom the impact of severe vibration and humidity.

2. According to the *Commitment in the Contract of Purchase on the Package and Transport Arrangements*, during the transport process it should be prevented from the impact of severe vibration and humidity with temperature at -20℃-55℃, relative humidity of not more than 93%/atmospheric pressure 500HPa~1060 HPa.

Chapter Ⅸ After-sale Service

Beijing TongRen Photo Electronic Corporation is responsible for the training, warranty, maintenance and technical support of the neurotransmitter tester.

I. Warranty

1. From the date on which the product is sold, any product quality problems will be repaired by the manufacturer.
2. During warranty period, it is free for maintenance and part replacement.
3. The duration of warranty is determined by the signed contract, and maintenance and technical service support is provided for life.
4. Products in the warranty period at the following two conditions are not covered in the warranty scope: abnormal use and their own demolition, to change equipment parameters and adjust the preset control and switch. Improper operation and random installation of programs other than from this device.
5. The warranty duration is 1 year.

II. Maintenance

1. Within 10 days of the sale, followup the customer bytelephone.
2. For products sold to the customers in the city, complete the maintenance within 24 hours.
3. When the products are sold to the customers in other cities need maintenance, perform telephone technical support first. For the ones need maintenance, send the customer turnaround machine to ensure the rate of running machines and then the customer send the fault machine back for maintenance.
4. Risks and damages caused by not following the specification ●risk precaution ● maintenance and curing should be on the account of the customer.

III. Maintenance Management

1. Setup files used by the customers
2. The after-sales service department is responsible for the maintenance and technical support.
3. The maintenance personnel must execute all the content in the Dispatch List.
4. Maintenance service hotline: +86-010-88203582
5. Complaint hotline: +86-010-88204236
6. Fax: +86-010-68236233

What determines who you are?

Surprisingly, six mysterious chemical transmittersin the brain, GABA, GLU, 5-HT, Ach, DA and NE, float to the surface of the water like an iceberg, of which the balance and imbalance determine whether you are healthy or ill,control your awareness, mentality, character and action,and fundamentally reflect the quality of your life.

How to get those 6 mysterious transmitters?

Brain function analyzer ET (Encephal of luctuographTechinology System) starts to reveal the effect of neurotransmitters on human health.